GOOGLE CLOUD × GEMINI ENTERPRISE AGENT PLATFORM

More or Less.

Architecture Brief — Snaptapp's first title

A real-time multiplayer trivia game where every generated answer is **grounded in Google Search**, calibrated by a **deterministic math kernel**, and audited before it reaches the table.

10	<2 s	0.0%	7
PLAYERS PER ROOM	GENERATED ROUND	ANSWER-LEAK GATE	METRICS BEFORE PROMOTION

THE SHORT VERSION

8 facts

- **One call per round.** Players decide whether the true figure is higher or lower than a believable decoy before the shot clock expires.
- **Grounded by default.** Custom topics call Gemini 3.5 Flash Lite with mandatory Google Search grounding; curated rounds serve from a verified fact bank.
- **Receipts, not promises.** Every generated round carries a `grounding_receipt` recording its web queries and support spans.
- **The model never picks the decoy.** DDM-4, a deterministic kernel shared by Go and TypeScript, derives every distractor number.
- **Audited before promotion.** Gemini 3.8 Flash cross-checks claims against primary sources; a 7-metric, 0.0% redline gate admits facts to Firestore.
- **Disputes get a referee.** The Fact Ombudsman runs adversarial Search checks on Cloud Run and evicts debunked facts in real time.
- **Fine-tuned on its own facts.** Audited facts export into Gemini SFT corpora and Gemma 2 9B LoRA adapters.
- **Built for rooms.** Up to 10 players sync on Durable Object WebSockets with a sub-20 ms shot clock.

LIVE **moreorless.snaptapp.us**
Play as a guest — no account required.

STUDIO

LEGAL ENTITY
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PRODUCT

TITLE
More or Less · live on web

PLATFORMS
Web now · iOS & Android in development

PLAYERS
1–10 per room · sub-20 ms sync

PRICE
Free to play · 5 live modes

PLATFORM

CLOUD
Google Cloud · `more-or-less-505908`

AI PLATFORM
Gemini Enterprise Agent Platform

MODELS
Gemini 3.5 Flash Lite · Gemini 3.8 Flash · Gemma 2 9B

GROUNDING
Google Search · mandatory on generated rounds

GUARDRAILS
DDM-4 math kernel · 7-metric gate

RUNTIME
Cloud Run (Go) · Cloud Firestore · Firebase

Inside 01 Hot path · 02 Verification · 03 The game · 04 Infrastructure

01 THE HOT PATH

Anatomy of a grounded round

From typed topic to synchronized answer in under two seconds — with a chain of custody a partner can audit. **Nothing about the number is left to the model.**

01
Scope gate
Topic text is filtered for scope and purity before any model call or quota is spent.

`pre-quota`

02
Cloud Run
`agent-go` receives an OIDC-authenticated request. No model keys live in the client.

`Google Cloud Run · OIDC`

03
Grounded generation
Gemini 3.5 Flash Lite answers with mandatory `google_search` grounding.

`Gemini 3.5 Flash Lite`

04
Deterministic decoy
DDM-4 derives the distractor from the verified claim across surprise factors 0–3.

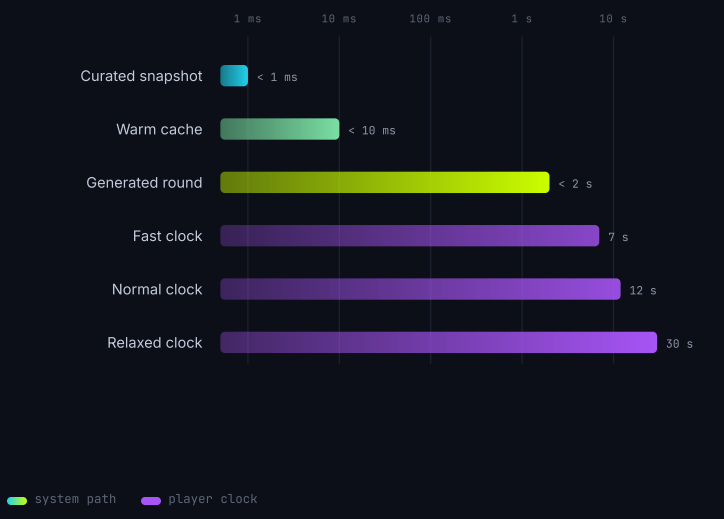
`DDM-4 kernel`

05
Stream & sync
NDJSON rounds stream to a Durable Object room; ten players share one shot clock.

`under 2 s`

Latency ladder

LOG SCALE · DOCUMENTED BUDGETS



Inside the receipt

PER GENERATED ROUND

`grounding_receipt` telemetry

- ✓ Model and grounding tool recorded
- ✓ Every web query retained
- ✓ Support spans mapped to the claim
- ✓ Decoy hash tied to the verified number
- ✓ Emitted before the round is served

MODEL	gemini-3.5-flash-lite
TOOL	google_search
GUARD	DDM-4 · SHA-256 digest

No receipt, no round. · auditable downstream

The round the player sees

EXAMPLE · ISS ORBITAL VELOCITY

CLAIM
Orbital velocity of the International Space Station
7,000.0 MPH
MORE | LESS

→

CALL
MORE
12 s shot clock
5 rounds · Standard

→

REVEAL
17,500.0 MPH
`grounding_receipt ✓`
true number · points · short explanation · report control

02 COLD PATH

The fact supply chain

Discovery runs offline and feeds the hot path. A claim reaches a player only after it survives verification, calibration, and the promotion gate.

01 Crawl
Gemini 3.8 Flash expands a subject into candidate numeric claims and their sources.

02 Verify
Each claim is cross-checked against primary web sources with Google Search grounding.

03 Classify
Quantity kinds are resolved so units compare correctly before any distractor exists.

04 Calibrate
DDM-4 precomputes decoy sweeps across surprise factors `0-3` for every fact.

05 Gate
Seven metrics, zero tolerance: **0.0% redline** and **0.0% answer-leak**.

06 Promote
Facts move `pending → active` in Cloud Firestore, ready for the edge snapshot.

Model allocation

GEMINI ENTERPRISE AGENT PLATFORM

WORKLOAD	3.5 FLASH LITE	3.8 FLASH	GEMMA 2 9B	GOOGLE SEARCH
Live pack generation	●	○	○	●
Fact Foundry audit	○	●	○	●
Dispute referee	●	●	○	●
Ops copilots	○	●	○	○
SFT & adapters	○	●	●	○

● active role · ○ not used on that workload

PROMOTION GATE

7	0.0%	0.0%
metrics in the gate	redline rate	answer-leak rate

`pending` → `active` · one bad metric keeps a fact out

CLOSED-LOOP FINE-TUNING

audited facts → adapters → back to generation

01 Audited facts
Cloud Firestore · fact_bank

→

02 Hermetic export
agent-go export-sft

→

03 Fine-tune
Gemini SFT · Gemma 2 9B LoRA

→

04 Evaluate
7 metrics · 0.0% redline · model_adapters

When a player disputes a fact

FACT OMBUDSMAN · CLOUD RUN · ADVERSARIAL SEARCH

A player report opens an adversarial review: counter-evidence queries run against primary sources, then the Ombudsman issues one of three rulings and acts on it immediately.

VERIFIED_CANONICAL
Claim upheld. The fact stays active and the dispute closes.
→ remains in rotation

DEBUNKED_AND_EVICTED
Counter-evidence wins. The fact is removed and its cache entry evicted.
→ cached rounds invalidated

AMBIGUOUS_UNIT
Units or framing can't be reconciled. The fact is quarantined for review.
→ held out of rotation

03 THE PRODUCT

Five ways to play, one verified fact bank

Same grounded round format, different pressure. Curated modes read the verified snapshot; custom topics stream live through the hot path.

Standard
CORE
Five rounds on any topic you pick, against the clock you pick.
5 rounds · 7 / 12 / 30 s

Cutthroat
ENDLESS
An endless sudden-death run drawn from the verified fact bank.
one wrong answer ends it

Daily Challenge
SHARED
The same verified rounds for everyone each UTC day.
streaks · fixed timer

Versus 1v1
DUO
The same rounds, with live round-by-round progress.
head-to-head · Firestore

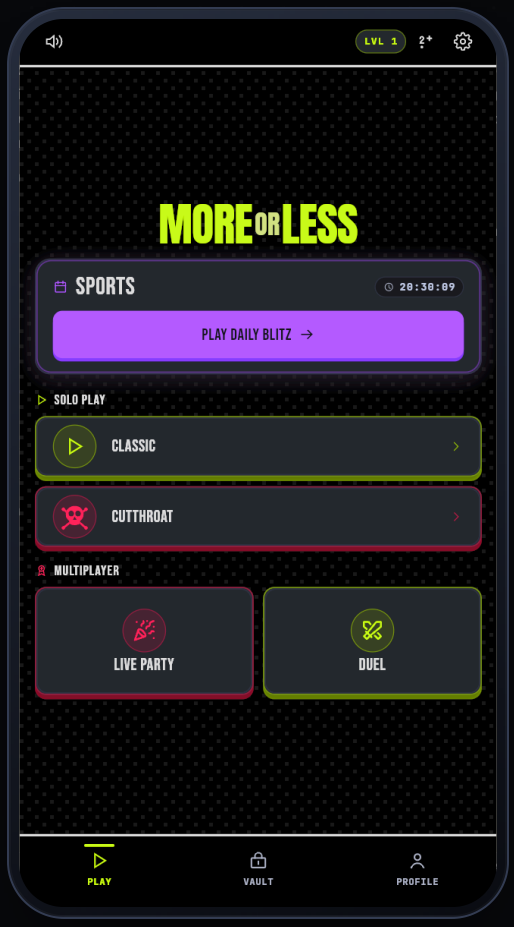
Party Room
MAX 10
Up to ten players share a room and a synchronized clock.
sub-20 ms WebSocket sync

60-Second Blitz
SOON
Rapid-fire 15-question daily time attack.
in development · mobile

Shot clock
chosen before the game

7	12	30
FAST	NORMAL	RELAXED
seconds	seconds	seconds

CUSTOM TOPICS — Type any subject. It is filtered for scope, generated on demand, and cached for other players who enter the same topic.



More or Less · web client

1–10 players per room
sub-20 ms sync



Party Room
Ten players, one room, one synchronized clock. Room state lives in a Durable Object.
sub-20 ms · no host advantage

One decision. Two buttons.
A verifiable answer every time.

MORE OR LESS
SHOT CLOCK 7 / 12 / 30 S

WEB LIVE · IOS IN DEVELOPMENT · ANDROID IN DEVELOPMENT · FREE TO PLAY

04 INFRASTRUCTURE

What we run on

Serverless by default, grounded by design. Managed Google Cloud and Firebase services keep a small studio operating globally with predictable economics.

Google Cloud services in production

PROJECT MORE-OR-LESS-505908

CLIENT	Flutter Web · WASM/Impeller · iOS · Android · Firebase Auth
EDGE	Durable Object WebSockets · sub-20 ms room sync · Firebase Hosting
AI PLATFORM	Gemini Enterprise Agent Platform · Gemini 3.5 Flash Lite · Gemini 3.8 Flash · Gemma 2 9B LoRA · Google Search grounding
SERVICES	Cloud Run · agent-go · Cloud Run Jobs · foundry · Cloud Functions v2 · OIDC · OpenTelemetry
DATA	Cloud Firestore · fact_bank · Cloud Firestore · model_adapters · Firebase

Security posture

ATTESTATION ON EVERY CLIENT PATH

Firebase App Check
Attestation token attached to every client request.

Play Integrity
Android app and device integrity verdicts.

App Attest
iOS app and device attestation.

reCAPTCHA Enterprise
Web abuse and bot scoring.

GOOGLE-MANAGED

Cloud Run · Cloud Firestore · Firebase · Google Search · Gemini Enterprise Agent Platform

SNAPTAPP-BUILT

agent-go · DDM-4 · Fact Foundry · Fact Ombudsman · SFT export · Flutter client

DATA BOUNDARY

Canonical profiles, leaderboards, and server-authoritative match histories live in Cloud Firestore. Player answers are not used to train Snaptapp's own models.

24-month Google Cloud credit plan

FOUR PHASES · ONE STUDIO PIPELINE

MONTHS 0–6
Launch & scale
Live Gemini 3.5 Flash Lite inference and Search grounding concurrency across web, iOS, and Android launches.

MONTHS 6–12
Foundry throughput
Higher nightly crawl, audit, and dispute-arbitration volume with Gemini 3.8 Flash.

MONTHS 12–18
Continuous tuning
Gemini SFT and Gemma 2 9B evaluation jobs on a fixed cadence, behind the same 7-metric gate.

MONTHS 18–24
Studio scale
Cloud Run, Cloud Firestore, and Firebase across More or Less and Snaptapp's second title.

Reviewing the stack for a partnership or the Google for Startups Cloud Program?

The production project is more-or-less-505908. Happy to walk through any layer of the pipeline.

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SCAN FOR THE PRODUCT PAGE
snaptapp.us/products